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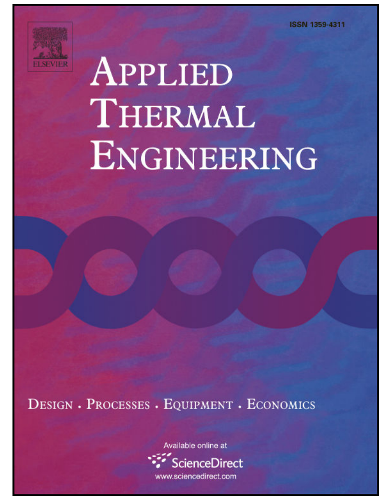

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Design, Optimization and Comparative Study of a Solar CPC with a Fully Illuminated Tubular Receiver and a Fin Inverted V-Shaped Receiver

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Abstract

This work presents the design, optimization, and a comparative study of a compound parabolic concentrator (CPC) using a tube as a receiver with and without a fin inverted V-shaped receiver (finned receiver). The study was mainly based on numerical analysis obtained by using a ray-tracing simulator in CPCs, in order to evaluate the amount of solar irradiation collected on four dates of the year. A north-south orientation was chosen, with and without a solar tracking system, at a fixed tilt angle of 32 and 0° for each case. The results showed that a CPC using a tubular receiver with a solar tracking system and a surface tilt of 32 and 0° reaches solar irradiation levels of 27.96 and 25.35 kWh/m^2, while a CPC with a finned receiver reaches 26.87 and 24.98 kWh/m^2, respectively. In the case of arrangements without a solar tracking system, the values reached were 7.03 and 5.60 kWh/m^2 for tubular receiver and 16.83 and 12.85 kWh/m^2 for the finned receiver, respectively. CPC with a tubular receiver proved to be more feasible when using a solar tracking system, in contrast to a CPC with a finned receiver, which had better feasibility in arrangements that do not include solar tracking.

Keywords: compound parabolic concentrator; fully illuminated tubular receiver; fin inverted V-shaped receiver; truncated solar concentrator; ray tracing analysis.

Nomenclature

Latin symbols

A	area (m^2)
A^*	parameter to describe coordinates of points on reflector
C	concentration
D	diameter (cm)
F_i	for (i =1, 2 and 3) is the focal point of the ith parabolic segment corresponding to the cavity
f	friction factor
f_{acc}	fraction of diffuse irradiation
f_i	for (i =1, 2 and 3) is the focal distance of the ith parabolic segment corresponding to the cavity (cm)
g_s	azimuth solar (°)
H	height (cm)
h_f	heat transfer fluid coefficient (W/m^2K)
h_s	solar altitude angle (°)
I_g	global solar radiation on the horizontal surface (W/m^2)
k	thermal conductivity (W/mK)
L	length (cm)
O	parabola edge
Pr	Prandtl number
P_1	center of the receiver base
P_1-P_2	circular arc describing the involute curve
P_2-P_3	first parabolic section
P_3-P_4	second parabolic section
P_4-P_5	third parabolic section
Re	Reynolds number
r	radius (cm); length of the receiver base (cm)
T	tangent to the receiver tube
V	velocity (m/s)
W	width (cm)
x	horizontal axis of the coordinate system
y	vertical axis of the coordinate system
z	deep axis if the coordinate system
ΔP	pressure drop (Pa)

Greek Symbols

β	tilt angle
$\boldsymbol{e_i}$	for (i =1, 2 and 3) is the axis of the ith parabolic segment
ε	absolute roughness (mm)
$\boldsymbol{\theta_a}$	acceptance half-angle (°)
θ_t	truncated acceptance half-angle (°)
θ_c	complementary angle (rad)
P	density (kg/m^3)
ϕ	angular parameter used to describe the geometry of the reflector surface (rad); latitude of the location (°)
γ	vertex angle of the receiver (°)

Subscripts

a	receiver
al	fin
cav	cavity
i	internal
r	reflector
t	truncated
opt	optimal

1. Introduction

The compound parabolic concentrator (CPC), also known as a non-imaging concentrator, was introduced for the first time by Winston [1] and subsequently by Baranov [2]. The CPC design consists of a parabolic and involute section on each side, which allows capturing of the direct and diffuse solar radiation that falls under a certain angle, to redirect it to a linear receiver. The advantages of this type of concentrator are they can function without a solar tracking system and still achieve some concentration. However, incorporate it can increase up 3.5 times the daily collected solar irradiation. The CPC could be used in a variety of solar applications, such as passive (integrated collector storage) and active (direct and indirect circulation) solar water heating, space heating and hot water production, heat pumps, sorption cooling and refrigeration systems, industrial air and water systems for process heat, desalination (multistage flash, multiple effect boiling, vapor compression), and solar chemical systems for thermal power systems [3].

Many authors have analyzed the geometrical and optical parameters of CPCs, seeking mainly to economize its design and maximize the concentration ratio. Rabl et al. [4] discussed various types of CPC solar collectors based on the absorber surface, such as flat, fin, inverted v-shaped (wedge), and tube absorber surfaces. McIntire [5] modified a small portion of the reflector, just below the absorber, creating a design with lower thermal losses. Baum and Gordon [6] developed analytic expressions for the collector height and reflector arc length of an ideal non-imaging collector with a cylindrical absorber for an arbitrary degree of reflector truncation. Baum and Gordon [7] investigated a non-imaging solar energy concentrator with a wedge-shape; the study developed an analytic solution for the minimization of the reflector, resulting in a reflector surface 4–11% smaller than a CPC with a plane absorber within a concentration interval of 1.2–2.0. Collares-Pereira [8] described and tested a non-evacuated CPC type concentrator with a "V" at the bottom, in order to reduce or eliminate optical losses due to the gap between the absorber and reflector. Gordon [9] derived analytic formulae for the geometric characteristics of a CPC with fully illuminated flat receivers, arranged horizontally or vertically. Khonkar and Sayigh [10] applied AutoCAD as an accurate ray tracing technique for the investigation of hot spots location on the tubular absorber of a CPC solar collector. They calculated the profile of the CPC and absorber, analyzed the intensity distribution around the circumference of the absorber, and analyzed the phenomena of rays inside the CPC at different incidence angles. Carvalho et al. [11] presented a CPC solar collector with an inverted “V” shaped absorber, as shown in Fig. 1a; the absorbers are not in contact with the reflector and the copper tube (located at the center) has two wings of thin aluminum plates, through which the heat transfer fluid passes. The gap produced between the tip of the mirrors (cusp) and the inverted V receiver leads to reduce optical losses, increasing the thermal performance of the CPC collector. Tripanagnostopoulos et al. [12] studied stationary asymmetric CPC collectors with flat bifacial absorbers; a flat bifacial absorber is installed at the upper part of the collector to form a thermal trap space between the reverse absorber surface and the circular part of the mirror. The experimental results showed that the proposed geometry could achieve a maximum efficiency of 0.71 with a multiunit collector. Adsten et al. [13] evaluated various asymmetric CPC designs with bi-facial absorber for stand-alone, roof, and wall-mounted installations, in order to achieve a higher temperature thermal energy. The annual energy outputs of the designs were 781, 925 and 490 MJ/m^2, respectively. Gata-Amaral et al. [14] presented the development of solar concentrating collectors based on complex parabolic collectors of solar energy, which incorporates a parabolic absorber with three cylindrical tubes, as shown in Fig. 1b. Tchinda [15] used a flat, one-sided absorber CPC solar collector for computing the thermal performance of an air heater. The efficiency obtained from the mathematical model showed good agreement with the experimental data, with an average error of 8%. Fraidenraich et al. [16] analyzed the class of concentrators satisfying the relation $\gamma \geq \theta_a$, as shown in Fig. 1c. The results showed that the ideal concentrator is a fully illuminated inverted V receiver, and incorporates on each side a circular involute plus three parabolic segments. Tíba and Fraidenraich [17] presented procedures for the geometric and optical optimization of the CPC type concentrator with a fully illuminated wedge absorber; the results showed that concentrators with a nominal concentration of 1.2 would save around 60% of the absorbent material of a flat-plane collector. They also established that in low concentration CPCs (C between 1 and 2), cavities with a minimal relationship between the length and height of the reflector surface and the aperture, (L/A) and (H/A), and the lower average number of reflections (n), correspond to the lowest angular acceptance concentrator (highest nominal concentration). Gu et al. [18] analyzed a non-tracking CPC solar collector with a flat double sided, horizontally positioned absorber. A ray-tracing optical analysis using experimentally measured properties was used to determine the solar flux distribution on the receiver as a function of the incidence angle (see Fig. 1d). Waghmare and Gulhane [19] analyzed the limiting diameter (LD) of the tubular receiver at the focus of the CPC using the geometrical method of ray tracing. They determined the elliptical shape, in which not a single reflected ray enters from the CPC half, and presented the geometric relation. Ustaoglu et al. [20] carried out a simulation using the ray tracing technique for the CPC solar collector, and found a significant improvement in the uniformity of the absorber, which can be achieved by truncating the reflector without substantial loss of performance. Wang et al. [21] proposed a model based on the edge ray principle, with a new V-shaped profile at the bottom of the reflector, and analyzed its parameters, such as the geometrical optic efficiency, transmittance, reflectance, and absorption ratio, with methods of ray tracing (TracePro) and geometrical optics. They concluded that a CPC with the new V-shaped profile has potential.

The research mentioned above has shown that the efficiency of a CPC in 2D can be improved by making changes to the receptor tube, without affecting its operability. Nevertheless, a detailed analysis of the geometry and operability of a CPC such as the one shown in Fig. 1e, which is a hybrid of a tubular receiver with a fin inverted V-shaped receiver, has not been completed. Thus, the present study evaluates the effects of making changes to the receptor tube, as well as the solar radiation collection and its operation during different dates of the year. The study is focused on a mathematical model and simulation using ray-tracing, in order to compare a tubular receiver with the proposed design.

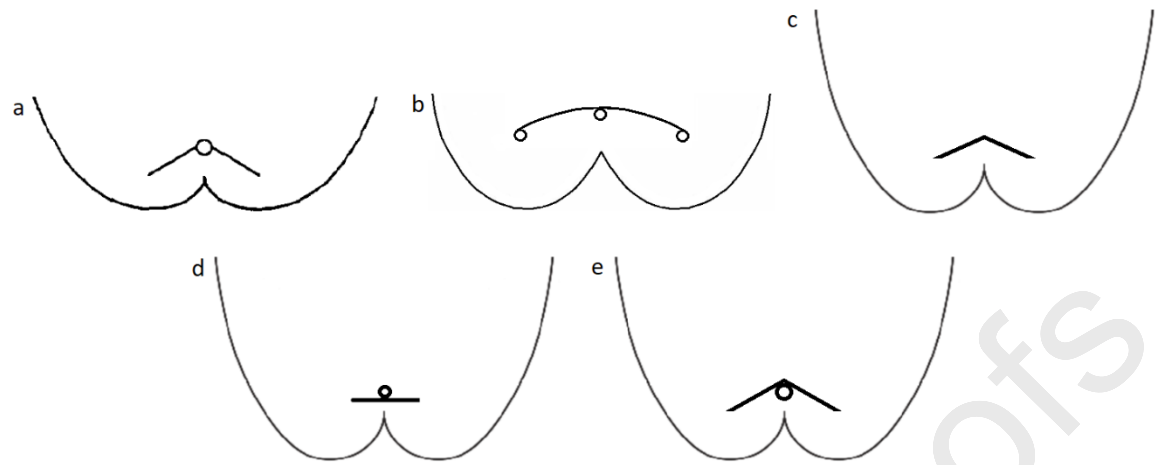

Fig. 1. Different configurations of finned receivers: a) inverted V shaped absorber, b) parabolic absorber, c) fully illuminated wedge absorber d) double-sided flat receiver, and e) tubular receiver and a fin inverted V-shaped receiver (proposal).

2. Materials and methods

The mathematical model calculated the geometry, and a parametric analysis was carried out to optimize the design. Afterwards, a software CAD was used to model the CPC geometric contours, which were optically analyzed with the purpose of quantifying the solar energy collected under different arrangements.

2.1. Geometric sizing

The design of the CPC reflector differs depending on the geometry of the receiver. In the case of a CPC with a tubular receiver, its design on each side employs a parabolic and involute symmetric section, while a CPC with a finned receiver has, on each side, an involute section and three parabolic sections, as shown in Figs. 1c and 1e.

A concentrator CPC with a tubular receiver is shown in Fig. 2, it comprised of a parabolic and involute symmetric section, a tubular receiver with diameter (D_a), and a concentrator width (W_{cav}). The acceptance half-angle (θ_a) defines the maximum concentration reached by the CPC, and it is obtained from its vertical axis, considering the ray path through the parabolic edge (O) and the tangent to the receiver tube (T). The equations used for the CPC design are listed in Table 1.

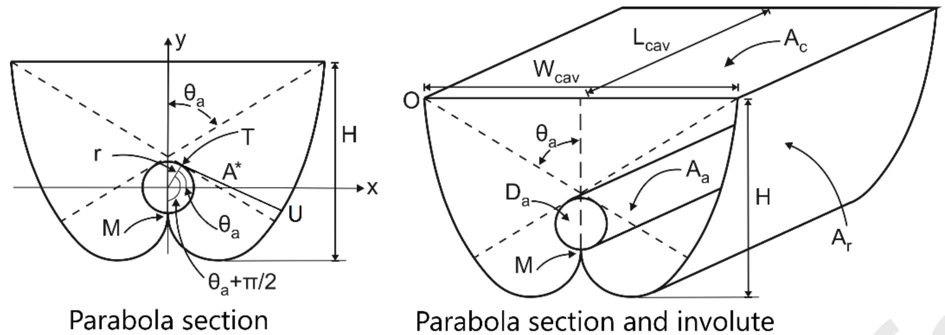

Fig. 2. Geometry of the CPC concentrator with a tubular receiver.

Table 1. Geometric equations of the CPC with a tubular receiver [22, 23].

Geometric concentration of the CPC	$C_{CPC} = \frac{1}{sin(\theta_a)}$	(1)
Equations for the parabolic section in the plane (*x*, *y*)	$x = r(sin\phi - A^* cos\phi)$ $y = -r(A^* sin\phi + cos\phi)$, $\frac{\pi}{2} + \theta_a \leq \phi \leq \frac{3\pi}{2} - \theta_a$	(2) (3)
Line from point *U* to the tangent of the receiver *T* is equal to the arc length MT along the receiver circumference	$A^* = \frac{\frac{\pi}{2} + \theta_a + \phi - cos(\phi - \theta_a)}{1 + sin(\phi - \theta_a)}$ Where for this point : $\phi = \theta_a + \frac{\pi}{2}$	(4)
Equations for the involute section in the plane (*x*, *y*)	$x = r(sin\phi - \phi cos\phi)$ $y = r(\phi sin\phi + cos\phi), 0 \leq \phi \leq \frac{\pi}{2} + \theta_a$	(5) (6)
CPC reflector area (proposed in this work)	$A_r = \left[\sum_{i=0}^{\frac{5}{2}\pi} \sqrt{(x^2 + y^2)}\right](L_{CPC})$	(7)
Tubular receiver area	$A_a = \pi D_a L_{CPC}$	(8)

A CPC with a tubular receiver and a fin inverted V-shaped receiver is shown in Fig. 3. This is shaped by a symmetric section of parabolas and involute, and a tubular receiver finned V-shape with a base length (*r*), tubular receiver with a diameter (D_a), vertex angle (γ), and concentrator width (W_{cav}). The acceptance half-angle (θ_a) is obtained from its vertical axis with respect to the ray path through the point (P_5) and focal point (F_3). Assuming that P_1 is defined as the central point of the receiver base and the CPC has its own axes (e_1, e_2, e_3) and focal points (F_1, F_2, F_3), it is possible to generate the first part of the reflector for tracing the involute curve based on the circular arc P_1-P_2 with an angle equal to $(\pi/2 + \theta_a)$. Consequently, the first parabolic section P_2-P_3 with focal point F_1 and axis e_1 can be generated when the continuity is given to the curve of the involute (point P_2) until the intersection with the straight line segment F_2-F_1 (point P_3). After that, the second parabolic section is created (P_3-P_4), which ends at the point of intersection with the straight line of segment F_2-F_3 (point P_4); this segment of the parabola has its focal point in F_2 and axis e_2, parallel to the axis e_1. Finally, the third parabolic section P_4-P_5 (with focal point F_3 and axis e_3) is guided into the intersection with an extreme radio passing through focal point F_3 with an angle θ_a (point P_5). Note that the variable (ϕ) corresponds to the angle measured counterclockwise for the involute with a center in F_1. The remaining parabolic section ϕ acts similarly to the parallel path or equal to the axis x from points F_1, F_2 and F_3. It is necessary to consider that if the variable γ becomes equal to θ_a, both parabolic sections P_2-P_3 and P_4-P_5 disappear, and the cavity would be shaped by an involute and the second parabolic section. Nevertheless, whether the receiver's fin becomes completely horizontal (γ=90˚), the second parabolic reflector section P_3-P_4 would disappear, causing the middle side of the concentrator to be shaped by an involute section and by the first and third reflector section [16].

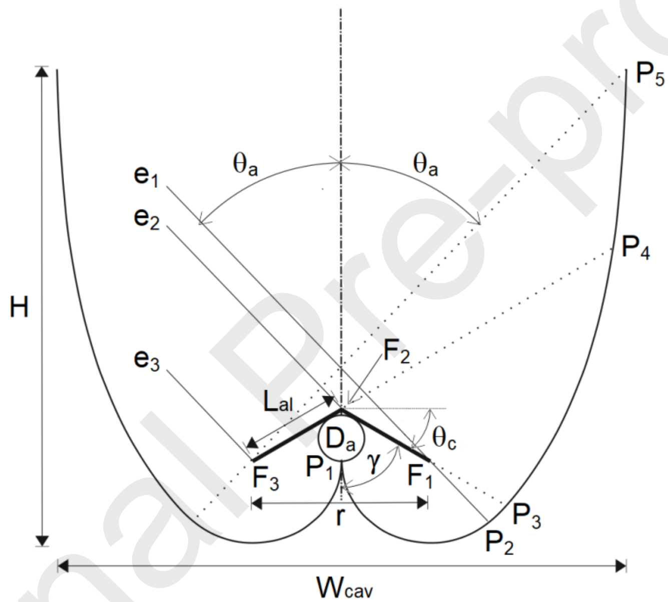

Fig. 3. Geometry of a CPC with a tubular receiver and a fin inverted V-shaped receiver.

Table 2 lists the geometric equations used to design the CPC concentrator based on the Cartesian coordinates system.

Table 2. Geometric equations for a CPC with a tubular receiver and a fin inverted V-shaped receiver [16, 24].

Geometric concentration of the CPC	$C_{CPC}=\frac{1}{sin(\theta_a)}$	(9)
Equations for setting the involute in the plane (*x*, *y*)	$x=\frac{r}{2}[1+cos\,(\phi)]$	(10)
	$y=\frac{r}{2}sin\,(\phi), \pi\leq\phi\leq 3\frac{\pi}{2}+\theta_a$	(11)
Equations for setting the first parabolic section in the plane (*x*, *y*)	$x=\frac{r}{2}\left[1+\frac{2cos\,(\phi)}{1+sin\,(\theta_a-\phi)}\right]$	(12)
	$y=\frac{rsin\,(\phi)}{1+sin\,(\theta_a-\phi)}, 3\frac{\pi}{2}+\theta_a\leq\phi\leq 3\frac{\pi}{2}+\gamma$	(13)
Focal distance corresponding to the first parabolic section	$f_1=\frac{r}{2}$	(14)
Equations for setting the second parabolic section in the plane (*x*, *y*)	$x=\frac{2f_2}{1+sin\,(\theta_a-\phi)}cos\,(\phi)$	(15)
	$y=\frac{rcos\,(\gamma)}{2sin\,(\gamma)}+\frac{2f_2}{1+sin\,(\theta_a-\phi)}sin\,(\phi),$ $3\frac{\pi}{2}+\gamma\leq\phi\leq 2\pi+\theta_c$	(16)
Focal distance corresponding to the second parabolic section	$f_2=\frac{r}{2}\left[1+\frac{1+cos\,(\gamma-\theta_a)}{2sin\,(\gamma)}\right]$	(17)
Equations for setting the third parabolic section in the plane (*x*, *y*)	$x=\frac{2f_3cos\,(\phi)}{1+sin\,(\theta_a-\phi)}-\frac{r}{2}$	(18)
	$y=\frac{2f_3sin\,(\phi)}{1+sin\,(\theta_a-\phi)}, \quad 2\pi+\theta_c\leq\phi\leq\frac{5}{2}\pi-\theta_a$	(19)
Focal distance corresponding to the third parabolic section	$f_3=f_2+\frac{r[1-cos\,(\gamma+\theta_a)]}{4sin\,(\gamma)}$	(20)
CPC reflector area (proposed in this work)	$A_r=\left[\sum_{i=0}^{\frac{5}{2}\pi}\sqrt{(x^2+y^2)}\right](L_{CPC})$	(21)
Receiver area	$A_a=[((\pi D_a)+(4L_{al}))L_{CPC}]$	(22)
Geometric concentration of the CPC	$C_{CPC}=\frac{1}{sin(\theta_a)}$	(23)

2.2. Parametric study

The geometric equations for both CPCs (tubular and finned) were programmed in Matlab® to carry out the parametric study. The mathematical model's input values were a width of 56 cm and a length of 2.44 m, which corresponds to the dimensions used in the construction of a CPC pilot plant installed in Mexicali, Mexico (Fig. 4).

Fig. 4. Pilot plant comprised of two CPCs with tubular receivers and two CPCs with tubular receivers and a fin inverted V-shaped receiver.

The permissible range of evaluations for a CPC with a tubular receiver was given by three parameters: receiver diameter, acceptance half-angle, and truncated height (H_t), since a geometry that incorporates a fixed receiver diameter can increase its width and height by decreasing the acceptance half-angle, however when this angle increases the opposite occurs; in either case, it is fundamental to use the truncation height to obtain the desired width. The parametric study of a CPC with a tubular receiver consisted of analyzing the commercial tubular diameters of 2.6, 3.3, 4.2, 4.8, 6.0, and 7.3 cm. Each diameter was evaluated with different acceptance angles and truncated heights to generate a design with an adequate concentration ratio.

On the other hand, for the CPC geometries with tubular receivers and a fin inverted V-shaped receiver, the permissible range of evaluations was given by two parameters: receiver diameter and vertex angle. The latter is defined by the gap available among the vertexes comprised by the edge-rays of the CPC geometry (F_2) and the center of the receiver base (P_1). The obtained maximum vertex angle corresponds to the angle formed by the V-shape fins, when the receiver is in direct contact with P_1, while the lowest vertex angle is obtained when the vertex angle is equal to the acceptance half-angle ($\gamma=\theta_a$). Different finned receivers were evaluated with diameters of commercial pipes under different acceptance angles, and afterwards each one of them was tested with different vertex angles in the permissible range until the desired geometric features were obtained. Finned geometry with the desired dimensions that can incorporate more than one diameter of tubular receiver between the underside of the fin and point P_1, the heat transfer fluid coefficient (h_f), and the pressure drop (ΔP) throughout these were calculated, as additional parameters, to select the diameter that provides the optimal thermal and fluid dynamics conditions. For the calculation of the heat transfer fluid coefficient in smooth tubes where the fluid is heated, the Dittus–Bolter correlation [25] is adequate:

$$h_f = 0.023Re^{0.8}Pr^{0.4}\frac{k}{D_i}, \quad (Re>10^4,\ \ 0.7<Pr<120,\ \ L_{cav}/D_i>10) \tag{24}$$

To obtain the pressure drop inside of pipes the Darcy-Weisbach equation [26] was used, along with the Colebrook-White equation [27] for fully turbulent flow:

$$\Delta P = f\frac{L_{cav}\rho V^2}{D_i\ \ 2} \tag{25}$$

$$\frac{1}{\sqrt{f}} = -2Log10\left[\frac{\varepsilon/D_i}{3.7} + \frac{2.51}{Re\sqrt{f}}\right] \tag{26}$$

Once the CPC geometries were defined, the contours were entered into the Solidworks® software to turn them into 3D objects, and then we imported them into the ray-tracing software, where the ideal optical properties of absorbance and reflectance were assigned to the receivers and reflective surfaces, respectively.

Regarding the establishment of ray grid, we based on an isotropic model of the diffuse irradiance that predicts the fraction of diffuse irradiation (f_{acc}) used by the CPC designs under study [28]. The general equation is a function of the CPC acceptance half-angle, CPC tilt angle (β) and the regular distribution of the diffuse radiance:

$$f_{acc} = \frac{2}{C_{CPC}[1+\cos(\beta)]} \qquad (\beta+\theta_a) \leq 90^\circ \tag{27}$$

$$= \frac{1+[C_{CPC}\cos(\beta)]}{C_{CPC}[1+\cos(\beta)]} \qquad (\beta+\theta_a) > 90^\circ$$

3. Results

The parametric study of a CPC with a tubular receiver consisted of six commercial tubular diameters. As a result of this analysis, tubular receivers with diameters of 2.6 and 3.3 cm were evaluated; nevertheless, they did not accomplish the set width of 56 cm. The diameters 4.2 and 4.8 cm were evaluated by employing different acceptance half-angles. It was observed that with the increment of the diameters, decreasing the width and height of the CPC enabled us to truncate the CPC to the desired width. In trying to find a balance between available reflective material, manufacturing cost, and concentration ratio, it was found that a height of 62.3 cm, a concentration of 2.23 cm, and a truncation of 70% (saving material) is adequate for a tubular receiver diameter of 4.2 cm, while a tubular receiver diameter of 4.8 cm was obtained at a height of 64.8 cm, a concentration of 2.27, and a truncation of 65.3%. If the percentages of truncation had not been considered, the values mentioned above would have changed and their heights would correspond to 1.29 and 1.25 m, respectively. The manufacturing cost would considerably increase. As for the diameters of 6.0 and 7.3 cm, the concentration ratio reached in geometries with a height similar to 65 cm were lower; therefore, the diameters were discarded. Finally, the selected diameter was 4.2 cm, due to the geometric design saving 0.12 m^2 of reflective material with respect to a diameter of 4.8 cm, assuming a minimal reduction of 2% in the concentration ratio.

In the case of a CPC a with tubular receiver and a fin inverted V-shaped receiver, the parametric study consisted of four types of commercial tubular diameters 2.1, 2.6, 3.3 and 4.2 cm, to set the permissible range in each case. Each finned geometry was evaluated under different acceptance half-angles of 20, 30, 31.5 and 35˚. Angles less than 20˚ generated heights of CPC above 93 cm, while values above 35˚ generated heights less than 54 cm, moving away in both cases from the 62.3 cm height obtained in the CPC with a tubular receiver. After evaluating the range θ_a from 20 to 35˚ and adding different vertex angles in the permissible range, it was found that at θ_a=31.5˚ with γ=50˚ the requirements of the same width and height of the CPC concentrator with a tubular receiver were satisfied, with a C_{CPC}=1.91 and a relation H/W_{cav}=1.11.

Subsequently, the heat transfer fluid coefficient and pressure drop of the working fluid (water) were evaluated in order to calculate the adequate diameter of the receiver tube, using an absolute roughness (ε) of 0.26 mm and an average temperature of 80˚ C. According to Fig. 5, the exploration began with the minimum commercially available diameter required according to the gap available for placement. The numerical results show that a diameter of 2.1 cm generates considerable values for the heat transfer fluid coefficient and pressure drop, however as the receiver diameter increases these same parameters decrease. Increasing the mass flow produced elevations of the heat transfer fluid coefficient, however, for the diameter of 2.1 cm, pressure drop occurs 4.5 times more with respect to at a diameter of 2.6 cm when the mass flow rate is 0.45 kg/s; as for the rest of the diameters, they generated a 3.6 and 4.2 times greater reduction as the dimensions of the diameters increased. In order to obtain a heat transfer fluid coefficient value that was as large as possible and a pressure drop that did not rise considerably during the increase in the mass flow inside the tube, with a relatively low acquisition cost, the diameter of 2.6 cm was selected.

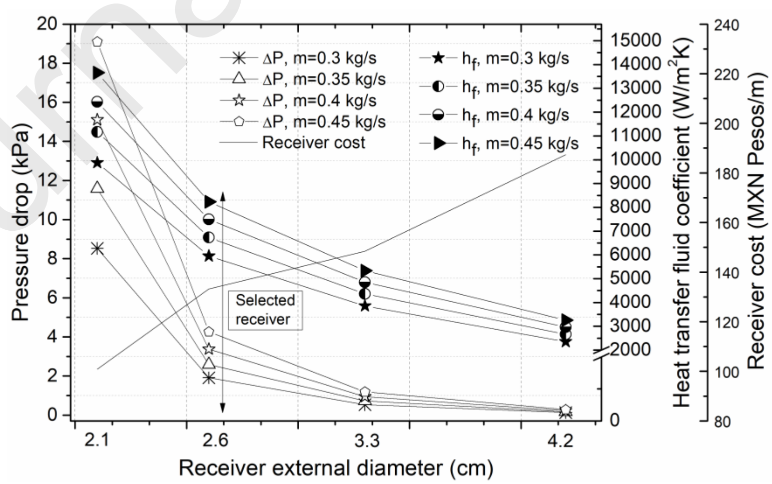

Fig. 5. Influence of receiver diameter on the heat transfer fluid coefficient and pressure drop.

Fig. 6 shows the CPC geometries at the same width and height. The most outstanding difference between both concentrators was the pronounced width across the edge of the CPC with the finned receiver, as well as the gap that the receiver presented concerning the central point of the base of the concentrator. Table 3 shows the final results of both developed geometries.

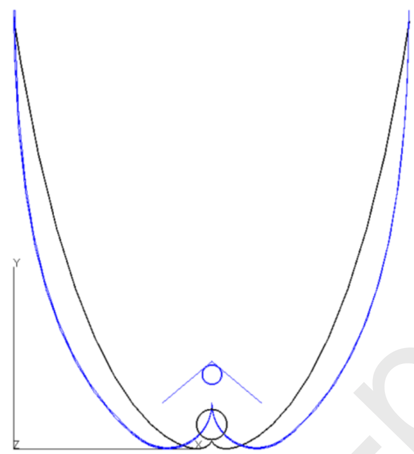

Fig. 6. Design of the CPC concentrators adjusted to the same width and height. Blue line: CPC with a tubular receiver and a fin inverted V-shaped receiver. Black line: CPC with a tubular receiver.

Table 3. Geometric parameters of the CPC designs

Geometric parameters	**CPC with tubular receiver**	**CPC with tubular receiver and a fin inverted V-shaped receiver**
Acceptance half-angle (θ_a)	-	31.5 °
Truncated acceptance half-angle (θ_t)	26.6 °	-
Width (W_{cav})	56 cm	56 cm
Height (H_t)	62.3 cm	62.3 cm
Length (L_{cav})	244 cm	244 cm
Percent truncated	70%	-
Concentration ratio (C_{CPC})	2.23	1.91
Vertex angle (γ)	-	50
Fin lenght (L_{al})	-	9.17 cm
Diameter of tubular receiver (D_a)	4.2 cm	2.6 cm

Fig. 7 shows the maximum and minimum fin ranges. It was found that it is possible to keep the receiver fully illuminated without having contact with the CPC reflective surface due to the heat transfer losses. Note that for certain values of θ_a, γ and r, a fixed width of the concentrator can be obtained as a result of an increment of the reflector height. Depending on the magnitudes of the variables, this will lead to a greater and lesser relationship between the height of the reflector surface and the aperture, (H/W_{cav}), which can be set to optimize designs to produce savings in reflective material and avoid a higher average value of reflection.

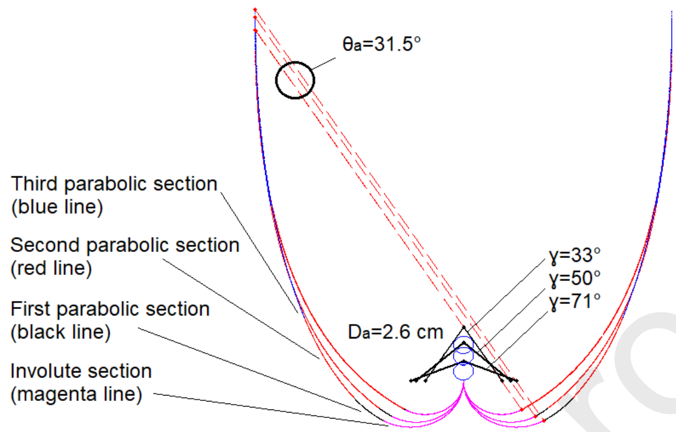

Fig. 7. Generation of the gap available for a finned receiver placement of 2.6 cm in the CPC geometry.

Ray-tracing software (TracePro®) was used to generate a series of simulations with the purpose of analyzing the reflective behavior of the incident solar radiation on the surface of the CPC geometries, and evaluating the collected energy. In this analysis, it was first necessary to specify the ray grid in the software, considering the relevance of the fraction of diffuse radiation. For arrangements of CPC concentrator with a tubular receiver without solar tracking systems with angle tilt of 0° and 32°, the fractions were 0.44 and 0.48, respectively, while for a CPC with a tubular receiver and a fin inverted V-shaped receiver were 0.52 and 0.56. Based on these values, it was established to carry out the ray tracing analyzes with global solar radiation. It is worth mentioning that any diffuse ray intercepted by the absorber, it does so with an angle less than or equal to θ_a, and for simplicity, it can be simulated as direct radiation. Values of global solar radiation were taken from the meteorological archive generated by Gallegos et al. [29], with data measured in the period 2000-2005 provided by the National Meteorological Service. The virtual geometries were placed in the space x, y, z to establish the solar incidence image of the geometries corresponding to the number of parallel rays, where each ray traced was equal to one watt of energy.

The ray-tracing study was performed considering the summer and winter solstices, as well as the spring and autumn equinoxes. In order to obtain the largest amount of energy, the CPC was evaluated with orientation north-south with tracking east-west under the four arrangements shown in Fig. 8. The time range used was from 8:00 AM to 4:00 PM, solar time.

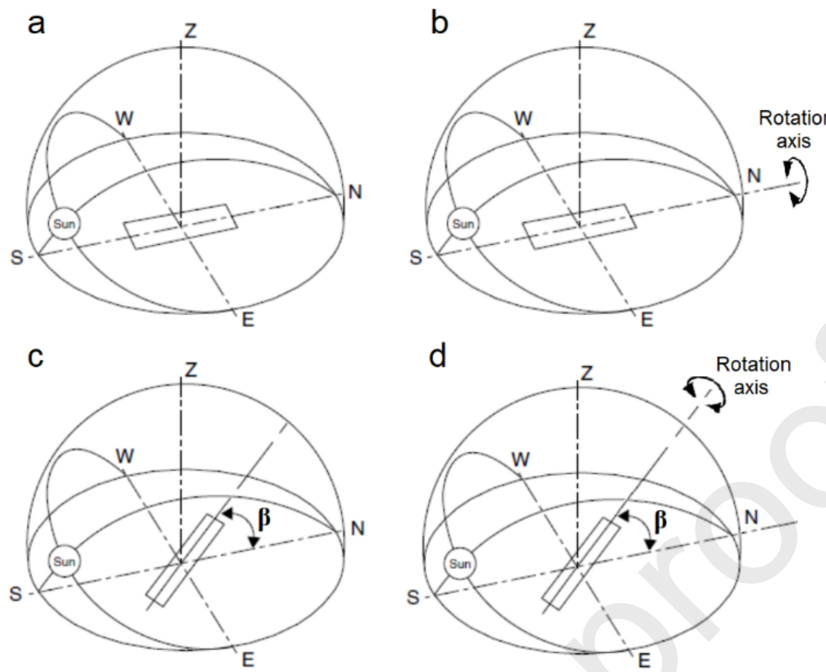

Fig. 8. Different arrangements of CPC solar tracking and tilt angle (β): a) without solar tracking-fixed 0˚, b) with solar tracking-fixed 0˚, c) without solar tracking-fixed 32˚ and d) with solar tracking-fixed 32˚.

Table 4 shows the parameters used in each hour evaluated in the ray-tracing software. The values correspond to the global solar radiation on the horizontal surface and were used to establish the ray grid that simulates the incident solar radiation on the plane. Regarding the location of the energy source (sun), the values corresponding to the solar altitude angle and solar azimuth angle were considered.

Table 4. Input parameter of the ray-tracing software.

Dates	Solar time	Solar altitude (h_s)˚	Solar azimuth (g_s)˚	Global solar radiation (I_g) W/m²
Spring equinox	08:00 a.m.	24.4	-72	282
	09:00 a.m.	36	-61	495
	10:00 a.m.	46.2	-46.2	681
	11:00 a.m.	53.7	-25.9	800
	12:00 p.m.	56.5	0	862
Summer solstice	08:00 a.m.	36.9	-96.3	686
	09:00 a.m.	49.5	-88.7	835
	10:00 a.m.	62.1	-78.5	920
	11:00 a.m.	73.9	-58.9	952
	12:00 p.m.	80.8	0	953

Autumn equinox	08:00 a.m.	24.3	-71.8	303
	09:00 a.m.	35.9	-60.7	515
	10:00 a.m.	46	-46	690
	11:00 a.m.	53	-25.7	813
	12:00 p.m.	56.3	0	847
Winter solstice	08:00 a.m.	9.9	-53.7	91
	09:00 a.m.	19.4	-43.3	263
	10:00 a.m.	27.1	-30.9	415
	11:00 a.m.	32.1	-16.1	522
	12:00 p.m.	33.9	0	574

Fig. 9 shows the reflective behavior of the CPC design in the ray-tracing software. The two CPC designs were evaluated independently, with its aperture exposed perpendicular to the incident solar image in such a way that the reflective surface redirected the solar energy collected and concentrated it. In this way, it was possible to quantify the amount of energy available to the receiver at a particular time and date.

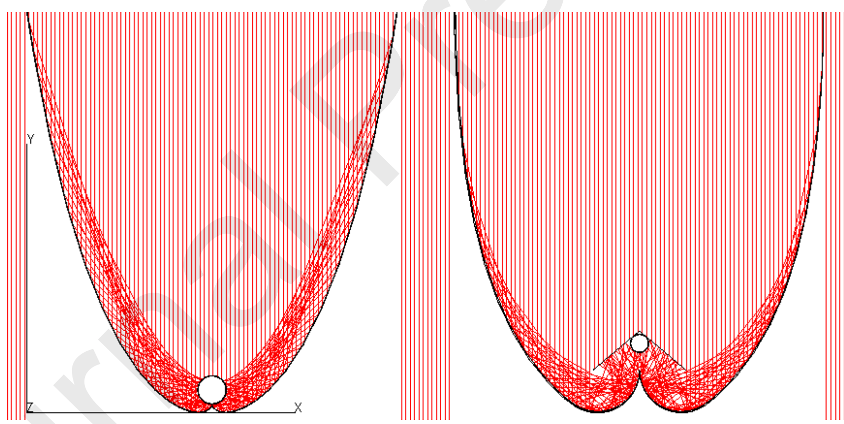

Fig. 9. Ray-tracing simulation of the CPCs at 12 pm solar time.

Fig. 10a shows the behavior of the daily solar energy collection corresponding to each CPC concentrator arrangement evaluated during the spring equinox. In the case of the two CPC arrangements with a solar tracking system, they harnessed the solar resource from 8 am to 4 pm due to the east-west movement, approaching 390 W/m^2 at the outermost parts of the day and a maximum of 1207 W/m^2 by 12 pm. The maximum solar energy collection throughout the day was achieved by the arrangement with a solar tracking system and tilt angle of 32˚, which obtained higher values than any other arrangement in the 9 hours of operation. In the case of the two arrangements without solar tracking systems, the number of hours of operation was affected when reduced from 10 am to 14 pm, because θ_t is relatively small by design (10.6˚), causing a reduction in its solar collection levels to 77.23%. It is important to note that this event was repeated for the rest of the evaluated dates, as it is a parameter that is inherent to the geometric design of the concentrator.

Fig. 10b shows the maximum solar energy collection evaluated during the summer solstice for 9 hours, which was obtained with the arrangement with a solar tracking system without tilt, followed by the arrangement with tracking and tilt. The difference in energy collection is due to the fact, that in the first arrangement, the rays are mostly perpendicular to the capture plane throughout the day, generating fewer energy losses at the ends of the concentrator. However, when the concentrator has a tilt angle of 32˚, these increase considerably. At 12 noon it was possible to observe a simultaneous decrease of the solar collection in both arrangements as a result of the same trend. In maximizing the value of the arrangement with tilt, it is recommended to use an optimum tilt angle of β_{opt}=17.6˚ to reduce the energy losses at the ends of the concentrator (longitudinal plane) by minimizing the incidence angle, which is directly affected by the variability of solar altitudes (see Table 4). The expression for an optimum tilt angle in the northern hemisphere ($\beta_{\mathrm{opt}}=\phi \pm 15˚$) is reported by Heywood [30] and depends on the latitude of the location ($\phi = 32.6˚$) and the season under study, where the plus, and minus signs are used in winter and summer respectively.

Fig. 10c shows that the autumn equinox presents trends and solar collection values similar to those obtained during the spring equinox, due to the implicit solar geometry and the global solar radiation levels on both dates. In Fig. 10d it is noted that the solar collection values obtained during the winter solstice are below those reached in previous dates. The solar energy collection values for arrangements that include a solar tracking system and a tilt angle of 32˚ are 715 W/m^2 at 12 pm and 108 W/m^2 for 8 and 16 pm, while for the arrangement with a tracking system and tilt angle of 0˚ its values are 395 W/m^2 and 94 W/m^2, respectively. This was because the tilt of the concentrator improves solar collection by 55.21%. In a concentrator with tilt, compared to one without tilt, the solar energy collection improves by 135.05%, since in the latter the solar collection is restricted until 10 am due to the acceptance half-angle it presents.

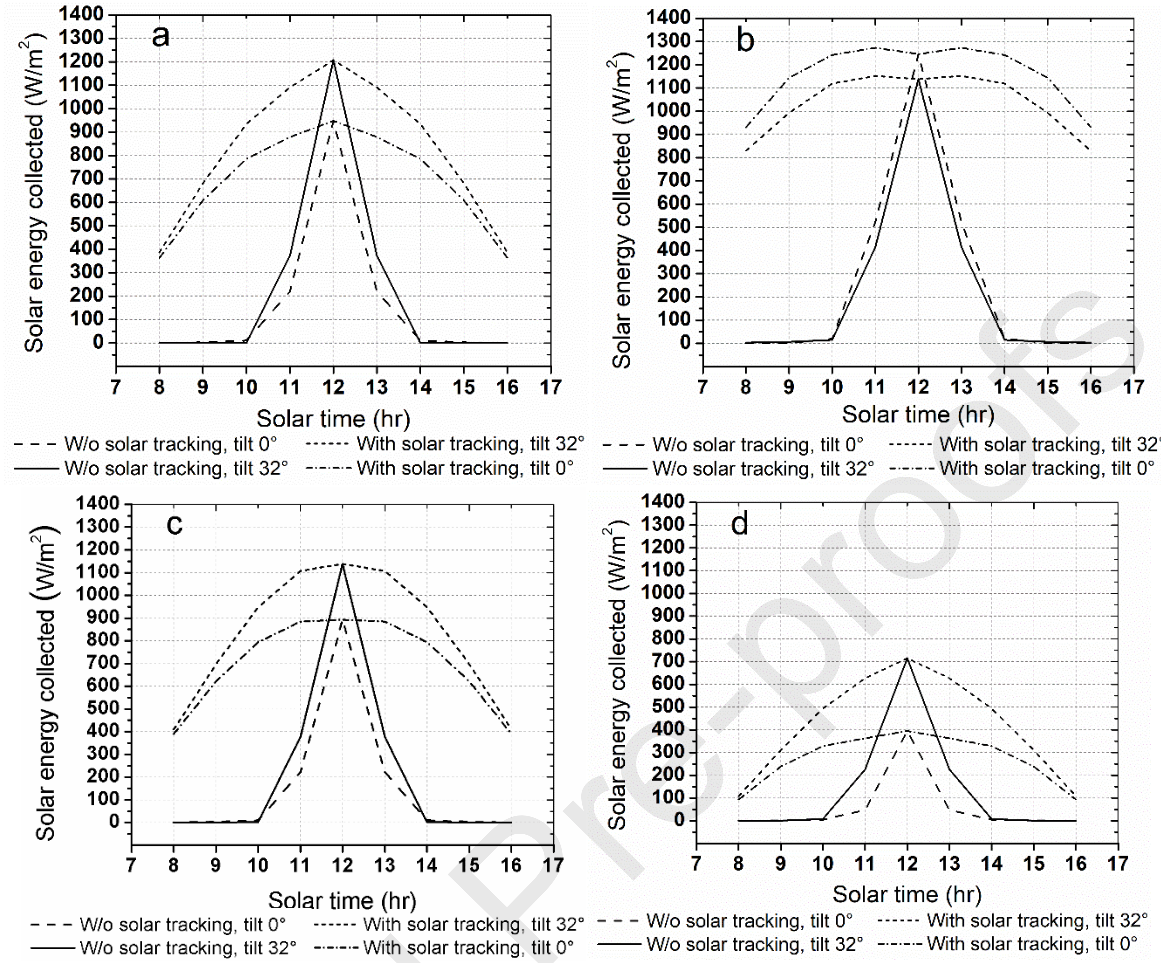

Fig. 10. Solar energy collected by CPC arrangements with a tubular receiver oriented north-south with and without a solar tracking system and different tilt angles: a) spring equinox, b) summer solstice, c) autumn equinox and d) winter solstice.

Fig. 11 shows the daily collected solar irradiation corresponding to each arrangement on the four dates analyzed. The maximum harnessing achieved by the arrangements corresponds to the solar tracking system with a tilt of 0˚ (10.42 kWh/m^2) during the summer solstice, followed by the arrangement with solar tracking and a tilt angle of 32˚. This difference in the energy collection is due to the fact that on that date, the sun reaches its maximum solar height during the 9 hours under evaluation, and the tilt used does not reach a sufficient perpendicularity with the capture plane when the tilt oscillates around 16˚. Note that for the following dates the arrangement with solar tracking and a tilt angle of 32˚ is beneficial when the value of the latitude of the place is used, which gives preference to an average solar collection during the year. Regarding the arrangements with a tilt of 0 and 32˚ that do not include solar tracking, they show the same behavior described above, since incorporating a tracking system helps to increase the solar irradiation collected in the first and last hours of the day.

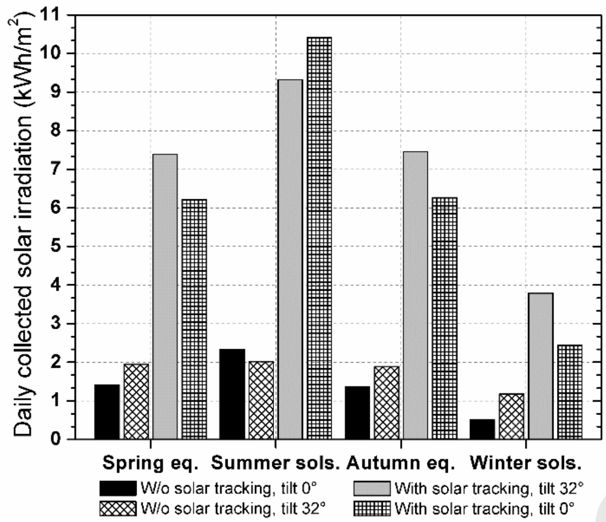

Fig. 11. Daily collected solar irradiation on different dates using a CPC with a tubular receiver.

Fig. 12 shows the accumulation of the solar irradiation collected from the different arrangements of CPC concentrators with tubular receivers evaluated during the four dates under study. The irradiation levels obtained by solar tracking arrangements with a tilt angle of 32 and 0˚ reached a values of 27.96 kWh/m^2 and 25.35 kWh/m^2, respectively, which indicates that tilting the concentrator causes the solar irradiation collection to increases by up to 10.30%. Solar irradiation collections of 7.03 kWh/m^2 and 5.60 kWh/m^2 were obtained with and without a tracking system, respectively. This represents an improvement of 25.54%. As result of this analysis, CPCs with a tilt angle of 32˚ that include a tracking system had better solar irradiation collection by up to 297.72% compared to those that do not include tracking. Similarly, a CPC with a tilt angle of 0˚ that included a tracking system increased collection by up to 352.68% compared to one that did not include it.

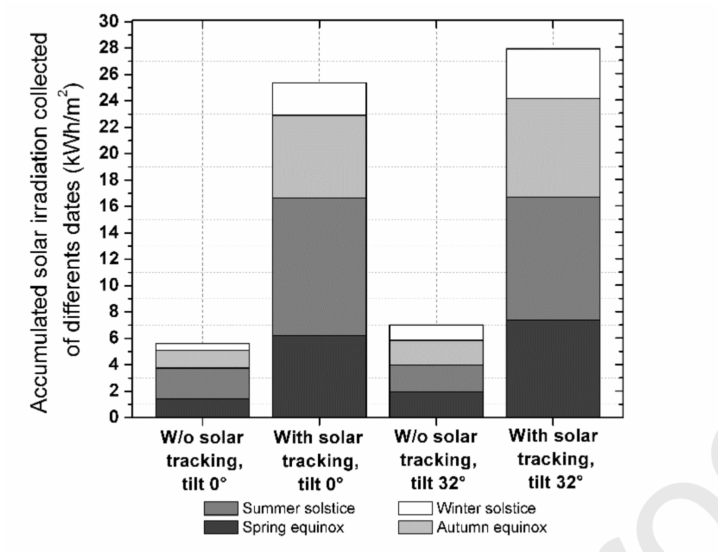

Fig. 12. Accumulated solar irradiation collected by the arrangements using a CPC with a tubular receiver.

On the other hand, Figs. 13a, 13b, 13c and 13d show the behavior of the daily solar energy collection using different arrangements of CPC tubular receivers and a fin inverted V-shaped receiver. The simulation conditions under which they were evaluated correspond to those used in the CPC arrangements with a tubular receiver. When the behavior was analyzed in the four figures, the same tendency was obtained in each of the arrangements shown in Figs. 10a, 10b, 10c and 10; however in the arrangements that do not have a tracking system, a higher range of operations is appreciated from 9 am to 15 pm, because the acceptance half-angle increased until 31.5˚ and this caused an increase in its collection levels to 48.35%.

In Fig. 13a, the two concentrator arrangements with a tracking system reached approaching 359 W/m^2 during the extremes of the day and a maximum of 1,140 W/m^2 at 12 pm from 8 am to 4 pm due to the east-west movement. The maximum solar energy collection throughout the day belongs to the arrangement with a solar tracking system and a tilt angle of 32˚, obtaining higher values compared to any arrangement in the 9 hours of operation.

Fig. 13b shows that a loss of energy at the ends of the concentrator is present in most of the arrangements, except for the arrangement that has a tracking system and tilt angle of 32˚. When the solar tracking arrangements of Fig. 10b are compared with those shown in Fig. 13b, these show greater energy losses at the ends, since by design the reflective surface of the CPC tubular receiver and a fin inverted V-shaped receiver become wider in the parabola section. In the arrangement with a tracking system and tilt angle of 32˚, as shown in Fig. 13b, the energy losses at the ends could be caused by the widening of the reflective surface, which is counteracted by the tilt of the solar concentrator. Therefore, this same widening is responsible for increasing the operating range, unlike in the CPC concentrator with the tubular receiver.

Figs. 13c and 13d shows that the behavior is the same as that described in Figs. 10c and 10d, however, in this case, the operating range becomes more visible, which will lead to greater daily collected solar irradiation, not only on the dates analyzed but on any day.

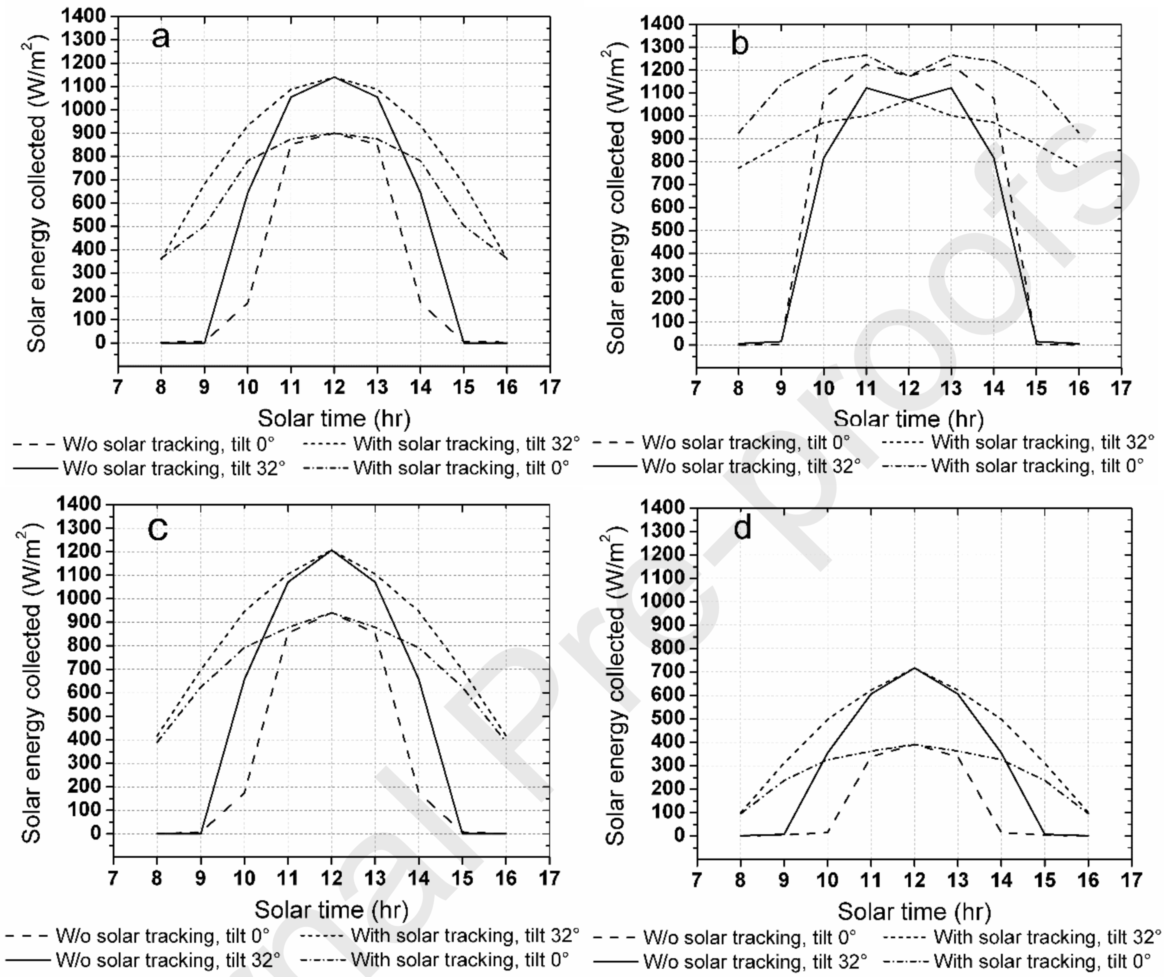

Fig. 13. Solar energy collected by CPC arrangements with tubular receivers and a fin inverted V-shaped receiver oriented north-south, with and without a solar tracking system and different tilt angles: a) spring equinox, b) summer solstice, c) autumn equinox and d) winter solstice.

Fig. 14 shows the daily collected solar irradiation corresponding to each arrangement on the four analyzed dates. Although the geometric design of the CPC tubular receiver and fin inverted V-shaped receiver presents slight changes inherent to its design with respect to the CPC concentrator with a tubular receiver, the maximum use achieved by the arrangements is retained by the arrangement with solar tracking system with a tilt of 0° in the solstice of summer, followed by the arrangement with solar tracking with a tilt of 32°. The following dates from this last arrangement are also beneficial when using the latitude value of the place. The arrangements with tilt angles of 0° and 32° without solar tracking have the same pattern described for the arrangements of CPC concentrators with a tubular receiver.

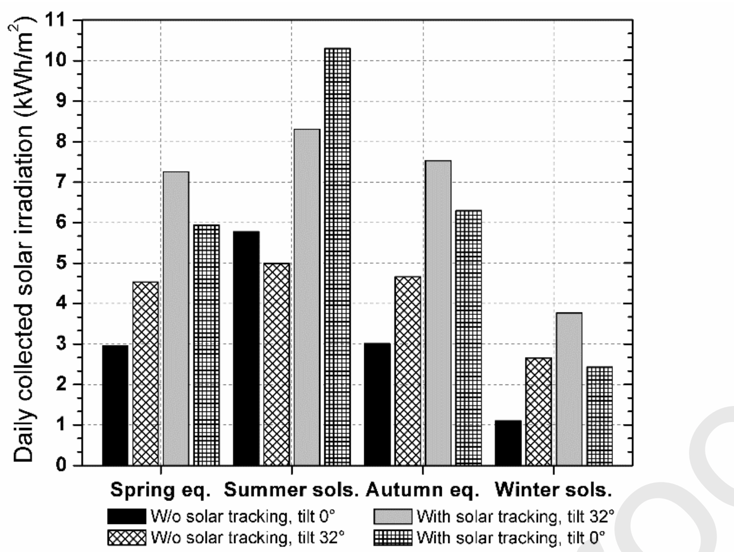

Fig. 14. Daily collected solar irradiation on different dates using a CPC with a tubular receiver and a fin inverted V-shaped receiver.

Fig. 15 shows the accumulation of the solar irradiation collected by the CPC tubular receiver and a fin inverted V-shaped receiver evaluated during the four dates under study. The irradiation levels obtained by the solar tracking arrangement with an angle of tilt of 32 and 0˚ reached a value of 26.87 kWh/m^2 and 24.98 kWh/m^2, respectively, which indicates that tilting the concentrator increases the solar irradiation collected by up to 7.57%. Comparing both tilt angles, the solar irradiation values without a tracking system, were 16.83 kWh/m^2 and 12.85 kWh/m^2, respectively; an increase of 30.97%. A concentrator with a tilt angle of 32˚ and a tracking system reached 26.87 kWh/m^2; the solar irradiance collected increased 59.65%, compared to the arrangement without a tracking system, which produced 16.83 kWh/m^2. Similarly, a concentrator with a tilt angle of 0˚ and a tracking system reached 24.98 kWh/m^2; the solar irradiance collected increased by 94.40% compared to the arrangement without a tracking system, which produced 12.85 kWh/m^2.

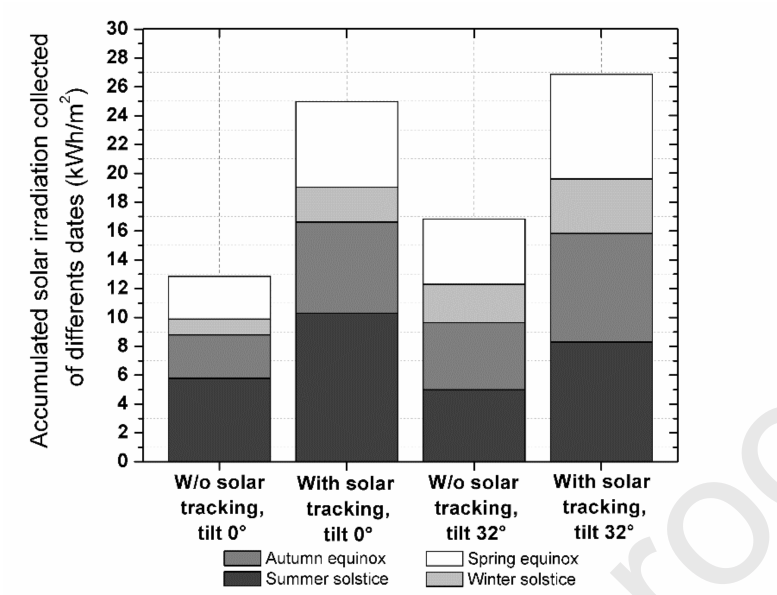

Fig. 15. Accumulated solar irradiation collected by arrangements using a CPC tubular receiver and a fin inverted V-shaped receiver.

Comparing the results shown in Figs. 10 and 15, the CPC concentrator with a tubular receiver and solar tracking system using an angle tilt of 32 and 0˚, with respect to the same CPC arrangements with a tubular receiver and a fin inverted V-shaped receiver, presented respective increases of 4.06% and 1.48% throughout the four dates under study. Similarly, a CPC tubular receiver and a fin inverted V-shaped receiver without a solar tracking system, using an angle tilt of 32 and 0˚, with respect to the same CPC arrangements with a tubular receiver, presented respective increases of 139.4% and 129.46%.

4. Conclusions

This work presented the design and optimization, as well as the results of a comparative study, of solar irradiation data collected from a compound parabolic concentrator (CPC) with a tubular receiver versus a CPC with a fully illuminated tubular receiver and a fin inverted V-shaped (finned receiver). The study was mainly based on numerical analysis obtained by using the ray-tracing simulator in both CPCs, in order to evaluate the amount of solar irradiation collected on four dates of the year. The CPCs were evaluated with a north-south orientation under the following scenarios: with and without solar a tracking system, with a fixed tilt angle of 32 and 0˚ for each case.

Both CPC designs were compared at a width of 56 cm and a height of 62.3 cm; the CPC with a tubular receiver resulted in an receiver diameter of 4.2 cm and a concentration ratio of 2.23, while the CPC with a tubular receiver and a fin inverted V-shaped resulted in an receiver diameter of 2.6 cm, vertex angle of 50˚, and a concentration ratio of 1.91.

Throughout the dates under study, the levels of solar irradiation obtained by the CPC arrangements with a tubular receiver and a solar tracking system with an angle tilt of 32 and 0˚ reached values of 27.96 and 25.35 kWh/m^2, respectively. In the case of arrangements without a solar tracking system, the values reached 7.03 and 5.60 kWh/m^2.

In the case of CPC concentrator with a finned receiver, the levels of solar irradiation obtained by the arrangements with a solar tracking system and an angle tilt of 32 and 0˚ reached values of 26.87 and 24.98 kWh/m^2, respectively, and in the case of arrangements without a solar tracking system, the values reached 16.83 and 12.85 kWh/m^2.

Additionally, it was established that arrangements of CPCs with tubular receivers and a solar tracking system using an angle tilt of 32 and 0˚, with respect to the same CPC arrangements with a tubular receiver and a fin inverted V-shaped, presented respective increases of 4.06% and 1.48% in solar irradiation collection, because the truncated acceptance half-angle of the tubular receiver was lower (26.6˚) than the half-angle acceptance of the finned receiver (31.5˚). Based on this, the CPC with a tubular receiver is viable in arrangements that incorporate a solar tracking system.

Regarding the arrangements of concentrators without solar tracking systems, it was found that a CPC with a finned receiver with an angle tilt of 32 and 0˚, with respect to the arrangements of CPC concentrators with a tubular receiver, presented respective increases of 139.40% and 129.46% in solar irradiation collection, since the half-angle acceptance of 31.5˚, together with the pronounced width across the edge of the CPC with the finned receiver, allowed an increase in the number of hours of operation, and decreased the energy losses at the ends of the concentrator. Accordingly, the CPC geometry with a finned receiver is viable in arrangements without a solar tracking system, unlike a CPC with a tubular receiver that, due to the narrowness of the geometry contour, causes greater energy losses at the ends, which can be compensated for with a solar tracking system and a higher concentration ratio.

Declaration of Competing Interest

The authors declared they have no conflicts of interest to this work.

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Highlights

- New variant of CPC with tubular receiver and a fin inverted V-shaped was analyzed
- Comparative study of ray-tracing between two geometries of CPC was carried out
- CPC with tubular receiver is viable in arrangements with solar tracking system
- CPC with finned receiver is viable in arrangements without solar tracking system
- Arrangements of CPC finned without solar tracking presents lower optical end losses